



**FIRST SEMESTER 2026-2027
COURSE HANDOUT**

Date: 30/07/2026

In addition to part I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course Code : ME G545
Name of the course : Deep Neural Networks
Instructor-in-charge : Madhurjya Dev Choudhury

I. Course Description:

Introduction to neural networks, approximation properties, backpropagation, deep network training, regularization and optimization, convolution neural networks, recurrent neural networks, attention models, transformers, neural architecture search, federated learning, meta-learning, applications in time series modeling and forecasting, online (incremental) learning.

II. Scope and Objective of the course:

This course is designed to impart the knowledge to understand the basics of deep learning and their applications in mechanical engineering. The objectives of this course are:

- ☐ Understand and explain the architecture, training, and optimization techniques used in deep neural networks.
- ☐ Apply deep learning techniques to solve problems in mechanical engineering domain.
- ☐ Analyze the performance and behavior of deep learning models.
- ☐ Evaluate the strengths and limitations of different deep learning architectures and techniques for mechanical engineering specific problems.
- ☐ Critically assess the latest advancements in deep learning.

III. Textbook and Reference Books:

- T1. Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola: Dive into Deep Learning, Cambridge University Press (2023).
- R1. Ian Goodfellow, Yoshua Bengio and Aaron Courville: Deep learning, MIT Press (2016).
- R2. Simon J. D. Prince: Understanding Deep Learning, MIT Press (2023).





IV. Course Contents

L. No.	Topics	Learning Objectives	Book (Chapter)
1-2	Introduction	Supervised learning, un-supervised learning, reinforcement learning, classification, regression	T1 (1), R2 (1,5)
3-5	Mathematical Foundation	Recap of basics of linear algebra, probability, information theory and optimization relevant to deep learning	T1 (2), R1 (2,3,4)
6-7	Coding	Introduction to coding, python, IDE, libraries	Lecture Notes
8-11	Neural Networks, Activation Functions	Neural networks for classification & regression, feedforward network, multi-layer perceptron, ReLu, Sigmoid, comparison of activation functions	T1 (3,4,5), R2(3,5), R1 (5, 6)
12-13	Optimization and Training	gradient descent, stochastic gradient descent, batch normalization, momentum, Adagrad, RMSprop, ADAM optimization, adaptive learning rates, overfitting and regularization, backpropagation	T1 (12), R1(5,6), R2(5,6,7)
14-16	Convolutional Neural Network (CNN)	Convolution, kernels, padding, stride, pooling	T1(7), R1(9), R2(10)
17-19	Deep CNN and Transfer Learning	AlexNet, VGG, ResNet, fine tuning	T1(8)
20-23	Generative Models	variational autoencoders, generative adversarial networks, diffusion models	R1(14, 20), R2(15,17,18)
24-26	Recurrent Neural Network (RNN)	RNN, sequence models, backpropagation with time	T1(9)
27-29	Modern Recurrent Neural Networks	Long short-term memory (LSTM), gated recurrent units (GRU), Bi-directional RNN, encoder-decoder architecture	T1(10)
30-32	Transformers	Queries, keys, values, attention mechanism, multi-head attention and self-attention, transformer architecture, vision architecture	T1 (11), R2(12)
33-34	Federated Learning	Introduction to federated learning, types and application in machinery fault diagnosis	Lecture Notes
35-38	Case-studies	Applications in smart manufacturing, machinery fault diagnosis and prognosis, additive manufacturing, fluid and thermal	Lecture Notes





V. Evaluation Scheme and Schedule:

Evaluation Components	Duration	Weightage (%)	Marks	Date and time	Remarks
Mid-semester Examination	90 min	25	25	<TEST_1>	CB
Project/Coding Assignment	-	30	30	To be announced in class	
Quiz	-	10	10		
Comprehensive Examination	180 min	35	35	<TEST_C>	CB
TOTAL			100		

VI. Chamber Consultation Hour: Thursday 2pm-3pm (Office: 1216-I).

VII. Notices concerning the course: All notices concerning the course will be communicated via email.

VIII. Make-up Policy: Make up will be permitted only in genuine cases with prior permission. Other than medical reason prior permission of IC is compulsory. No make up for quizzes.

IX. Minimum Passing Marks/NC: The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the post-comprehensive examination score OR 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course. NC will be awarded if a student does not appear in the mid-semester or comprehensive examination.

Instructor-in-Charge

ME G545

